APPROACH MEMO

Independent Cardiology Lead Engine

Nov 1, 2026 · Rohit Nistala

Summary

I built the first step of the funnel: finding hospital-employed cardiologists likely to go independent, roughly when, and which of them could start a group together. Everything else depends on that list.

On simulated data the top 25 leads move at twice the average rate, which mainly shows the code works. On real CMS data I scored 193 cardiologists using weights from a back-test (October 2022 to October 2024). Its top 50 held 10 movers (20%, against a 4.7% base rate), but it failed a second test, so I treat the real scores as a list to check, not a forecast. The next step is a 90-day pilot that produces real outcomes.

The problem

The Company mostly hears from cardiologists who already raised their hand. The bigger market, hospital-employed cardiologists, has to be found earlier. The brief lays out four steps, each depending on the one before:

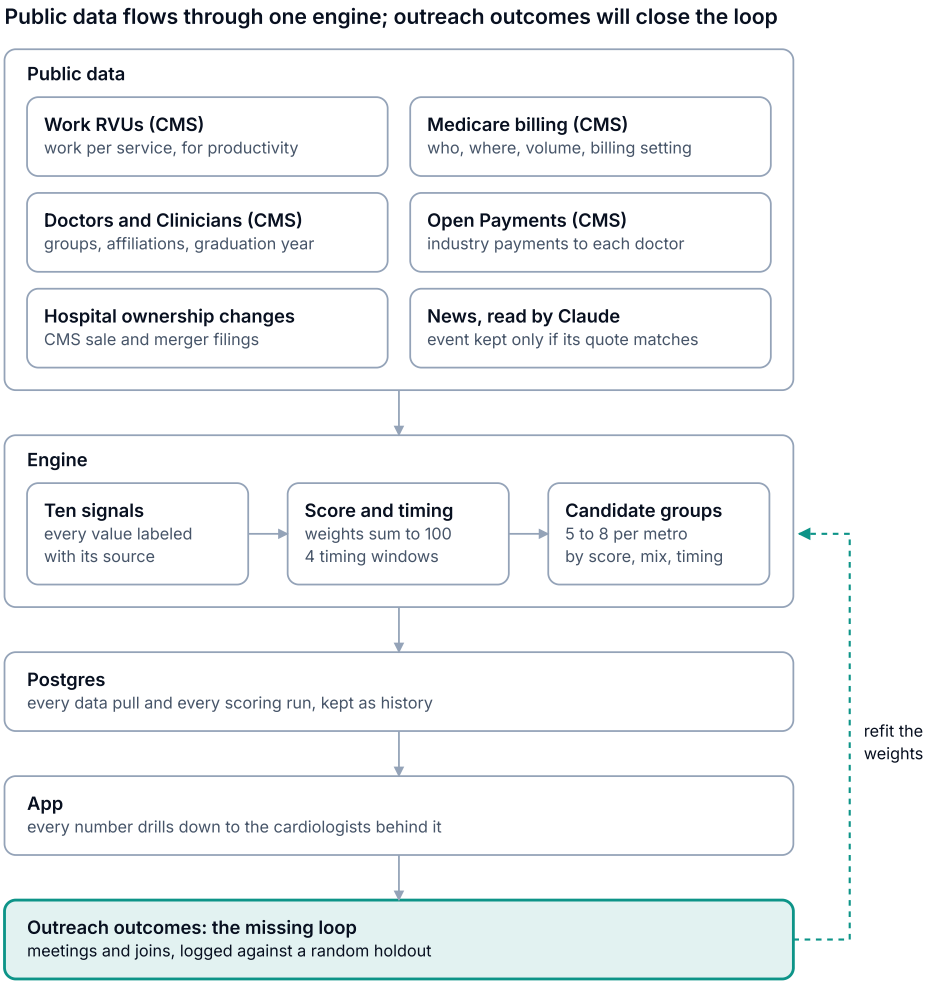
1. Work out who might move, and when.
2. Find and verify their contact details.
3. Check whether their interest is real.
4. Bring five or more together into a group.

I started with step 1, since nothing else works without it. Timing matters too: a call soon after a hospital merger or a colleague leaving should land better than a cold one.

My approach

Pick the step that unlocks the rest, build it end to end, and be clear about which numbers are real. Each cardiologist gets a score out of 100 made of points from ten signals, so every score comes with reasons a recruiter can repeat. Timing is a separate output.

Groups are built within each metro from doctors scoring 55 or more. Every possible slate gets a viability score out of 100: 40% average score, 25% size (up to eight), 20% subspecialty mix, and 15% the share whose window opens within nine months. The engine builds the strongest slate first by adding whichever doctor raises viability most, stops at eight, then repeats with who's left while at least five remain. Leftovers go on a bench, and the app shows how much each would add to an existing slate. The real data has no groups yet, because the score cut-off has no real basis there.



Pipeline: public data to the app, and the outcome loop the pilot adds

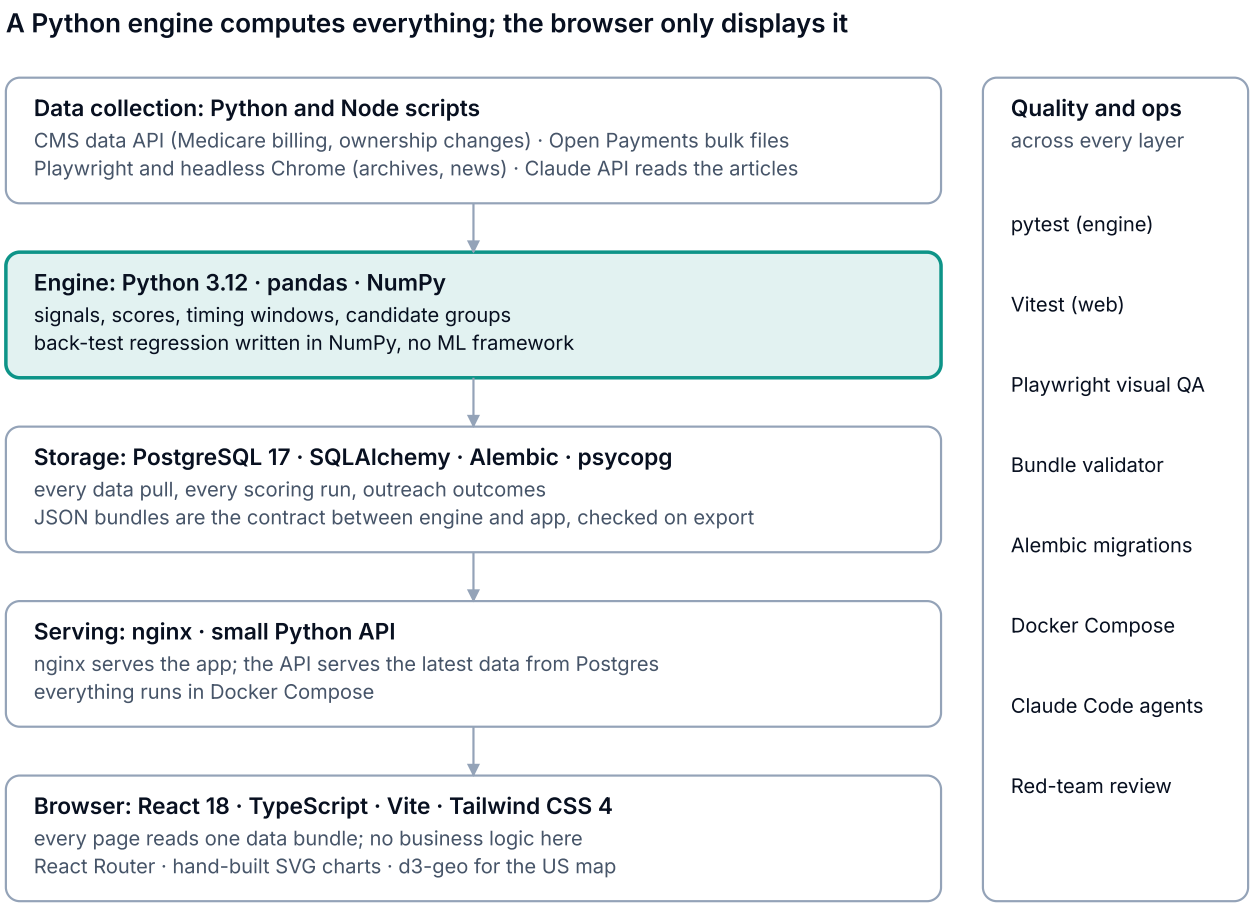
The dashed line is what we don't have yet: real outreach results feeding back into the weights.

What I built

MODE	WHO	SCORES	CHECKED AGAINST
Synthetic	1,200 simulated cardiologists, 8 metros	Scores, timing and candidate groups	A simulated outcome
Real data	205 hospital-based cardiologists in the same 8 cities, from CMS records	Scores for 193; no timing or groups yet	A 2022 to 2024 back-test

- Every location, cohort, doctor and signal has a page, and every number links to the people behind it.
- Claude reads news about each doctor's group and hospitals. An event only counts if its quote appears word for word in the article and it passes the signal's rules: 71 of 482 extracted events did.
- Postgres keeps every data pull and scoring run and serves the app in Docker.

Tech stack



Tech stack: from public data to the browser

The browser holds no business logic, so a change in the engine shows up everywhere in the app.

Results on simulated data

The top 25 leads moved at 44% against 20% overall, about twice as good as random. That mostly confirms the code, since the simulation shares the score's assumptions. It also shows group formation is hard: with 41% of qualified leads moving, a slate of eight yields about three founders.

Results on real data: the back-test

I rebuilt the signals as of October 2022 and checked who joined an independent-looking group by October 2024 (44 of 935, or 4.7%). A logistic regression, judged only on held-out doctors:

HELD-OUT RESULT	VALUE
Ranking quality (AUC, 0.5 = coin flip)	0.66, likely 0.57 to 0.75

HELD-OUT RESULT	VALUE
Top 50	10 movers (20%), 4.3 times the base rate
Top 100	14 movers (14%), 3.0 times
Second test: switched to office billing	No better than chance (0.55)

It passed the bar I set in advance, but the outcome is inferred, mover counts swing from 12 to 56 depending on the window, and it failed the second test. Six of nine signals also pointed the opposite way from what I assumed:

SIGNAL	POINTS	POINTS TOWARD MOVING	I ASSUMED
Productivity	24	Lower volume	Higher
Independent groups nearby	18	Fewer	More
Industry payments	16	More	More
Prior independent practice	13	Yes	Yes
Subspecialty economics	11	Less procedural	More
Colleague recently left	8	No	Yes
Recent employer change	5	No	Yes
Employer disruption	3	No	Yes
Career stage	2	Mid-career	Mid-career

Why every score is points

Recruiters need reasons they can repeat, so every score is a sum of points per signal. On real data each regression coefficient becomes its share of 100 points, so the ranking matches the model. With only 44 movers I'd keep the model simple until outreach data arrives.

When I'd move to other models

MODEL	WHEN	USED FOR
Claude reading news	In use	The disruption signal; later, call notes
Logistic regression	In use; refit on pilot outcomes	Points per signal
Survival model	Once moves have reliable dates	Timing windows
Gradient-boosted trees	Thousands of outcomes	Ranking with interactions

MODEL	WHEN	USED FOR
Sequence model (LSTM, RNN)	Years of monthly history	Early warning as events arrive
Interest classifier	A few hundred logged conversations	Ordering follow-ups

Outcomes come from CMS snapshots today (noisy) and from pilot results later (clean). I'd test a sequence model against a simple survival model before trusting it.

What I measure, what I don't, and what I'd add

METRIC	STATUS	WHY
Hit rate and lift in the top 25, 50 and 100	Tracked	Share of calls that reach a mover
Hit rate at each score cut-off	Tracked	Matches the list to team capacity
Each signal vs. chance, and a second outcome	Tracked	Catches weak signals and a score that fits only one outcome
Timing accuracy, calibration	Not tracked	No reliable move dates; the score isn't a probability
Contact accuracy, meetings, groups formed	Not tracked	No outreach yet
Timing error in months (MAE)	Next	Gap between the predicted window and the actual move
Calibration error (Brier score)	Next	Whether "30% likely" means 3 in 10
Hit rate vs. a random holdout	Next	Proves the score, not luck, made the difference
Conversion per stage, cost per group	Next	Where the funnel leaks

Timing error example: predicted four months out, left in seven, a three-month miss. MAE averages that across movers.

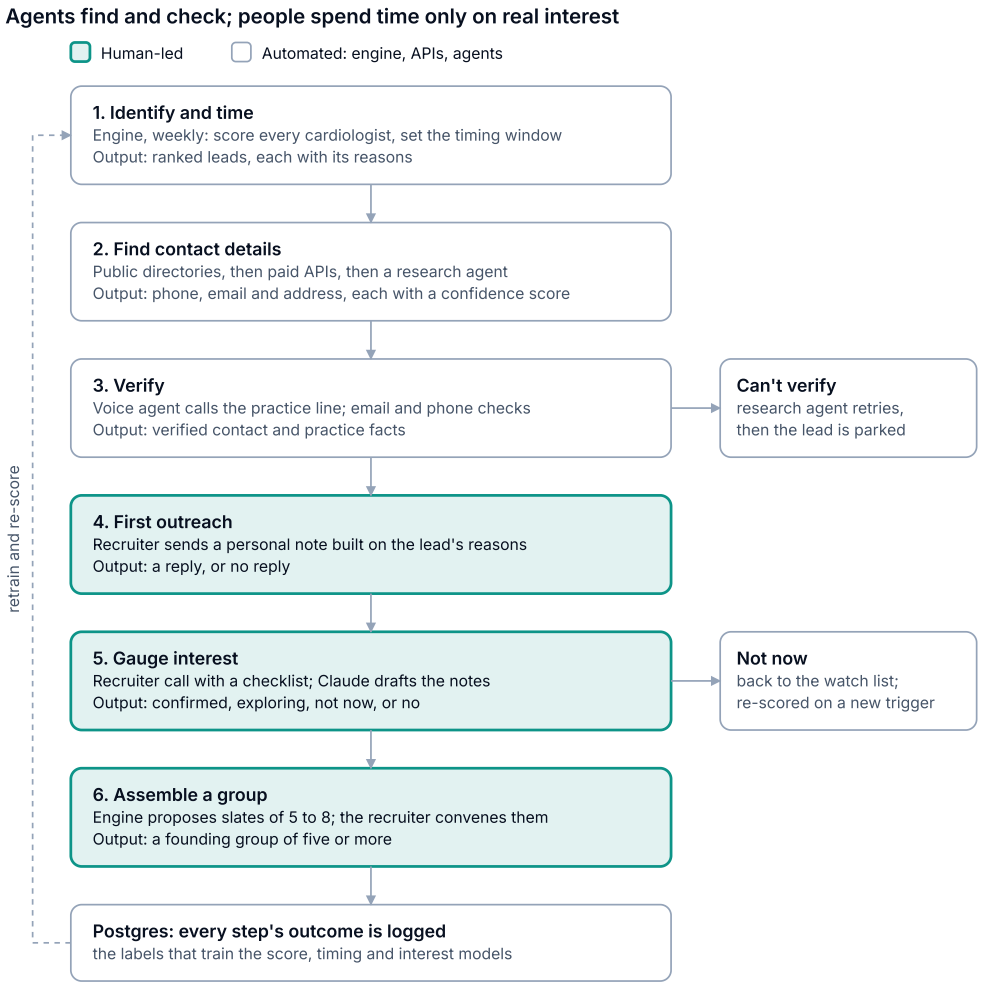
Signals I'd like but don't have

SIGNAL	WHERE IT WOULD COME FROM	WHY IT'S MISSING
Contract renewal date	Recruiter conversations	Not public
Non-compete terms	Conversations, state law	Not public
Actual employer	CMS enrollment records	Not connected yet
Hospital financial distress	Cost reports, bond disclosures	Not connected yet

SIGNAL	WHERE IT WOULD COME FROM	WHY IT'S MISSING
Pay against market	Conversations, salary surveys	Not public
Full patient volume	Commercial claims	Paid data
Who has worked with whom	CMS groups, shared patients in claims	Partly paid

The full workflow

Agents and APIs do the finding and checking, so recruiters only spend time on reachable, possibly interested doctors. I built step 1 and the group engine.



End-to-end workflow: six steps, three of them human-led

Finding and verifying contact details

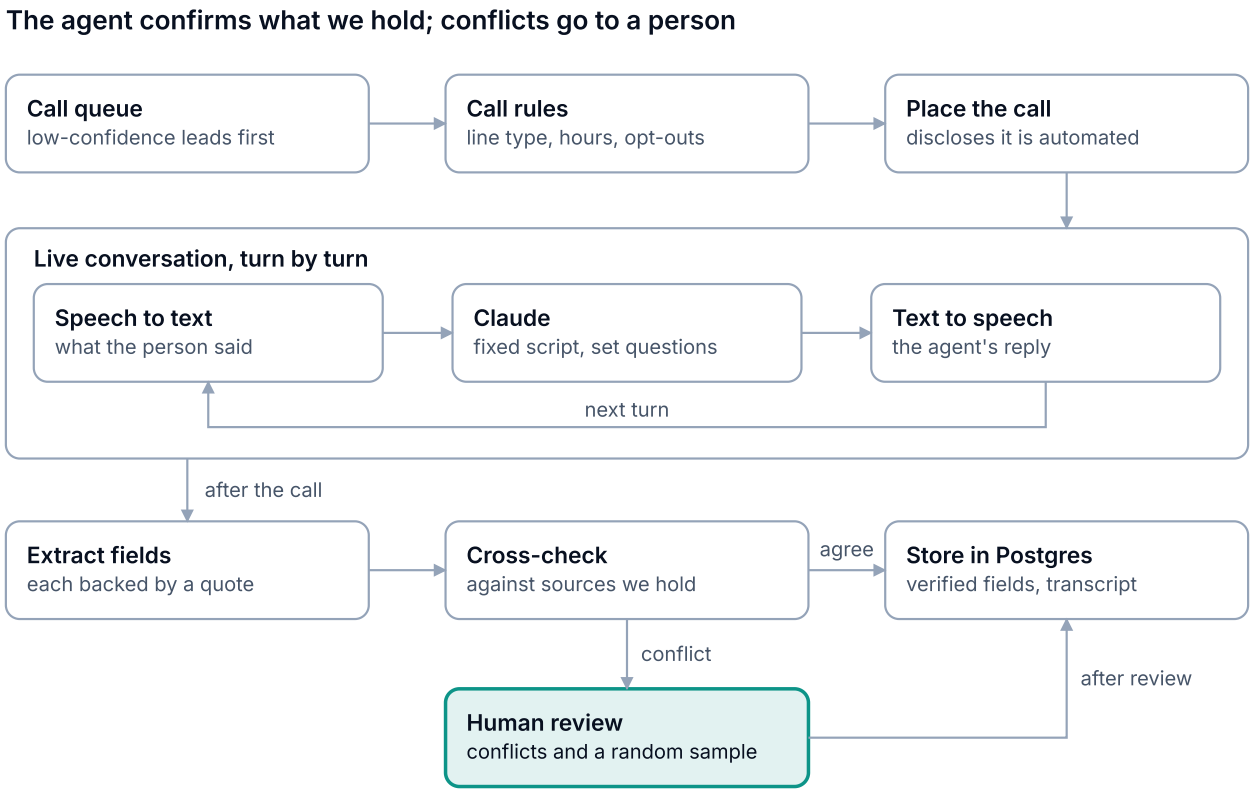
1. Free public directories first (NPI registry, CMS Doctors and Clinicians, practice websites).
2. Paid contact APIs for direct phones and emails, cheapest first.

3. Verification APIs to confirm emails accept mail and to tell mobiles from business lines.
4. A Claude research agent for gaps, accepted only when its source page contains the field.
5. Merge, score confidence, and send low-confidence contacts to the voice agent.

Email follows CAN-SPAM, calls follow the TCPA, and no patient information is collected.

Verifying doctor information with a voice agent

When we're unsure of a doctor's details, a voice agent calls the practice's main line to confirm them. It never pitches.



Voice agent: one call, from queue to verified record

It asks three fixed questions: is the doctor still at this location, which group is it part of, and how to reach them professionally. It discloses that it's automated, calls business lines only (the TCPA requires consent for AI voices on mobiles), honors opt-outs, and hands off to a person on request. I'd build on an existing voice platform and track connect rate, accuracy against human review, and cost per verified record.

Gauging whether interest is real

This stays human. Claude prepares a brief before the call and turns notes into a checklist after it.

QUESTION	REAL INTEREST LOOKS LIKE
Timeline	Names a month or event
Constraints	Has checked their non-compete
Economics	Asks about payers or income
Peers	Names colleagues who'd join
Next step	Agrees to a second meeting

The recruiter marks each doctor confirmed, exploring, not now, or no. Those labels feed the next model.

Trade-offs

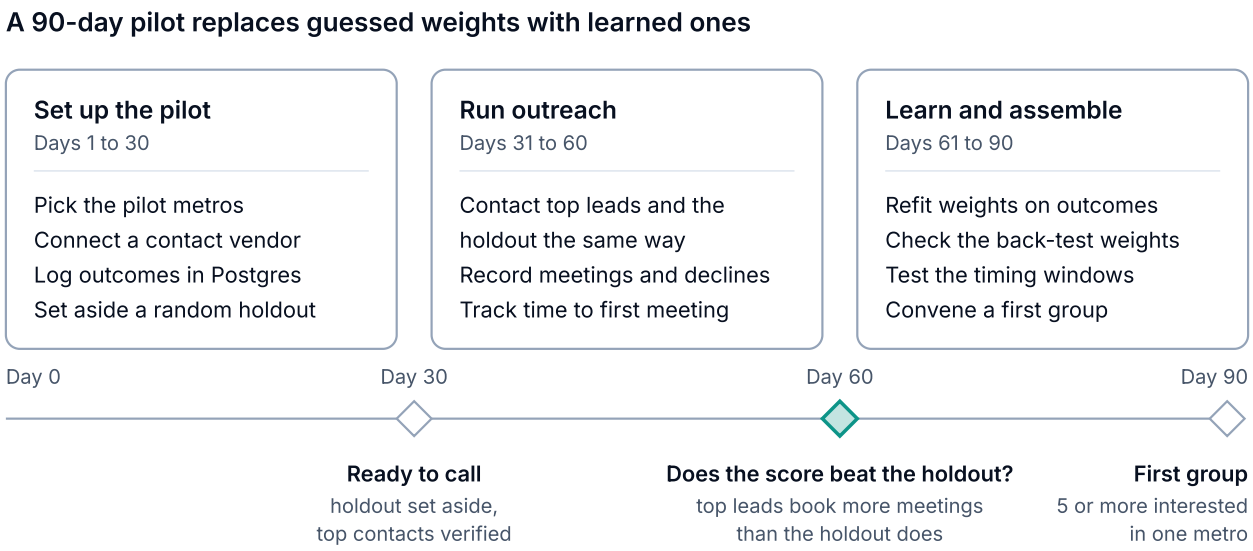
CHOICE	WHY	COST
Points per signal, not a black box	Recruiters need reasons	Rough weights for now
Built identification, not contact lookup	Vendors already solve contacts	List isn't call-ready yet
No automated outreach	Needs real conversations	No signal on interest yet
Scored real data from an inferred outcome	Tested beats untested	Can mislead if read as a forecast
Metro as the group unit	Simple and local	Misses groups across cities

If things change

IF	THEN
Outreach outcomes arrive	Refit the weights and compare against the holdout
A new data source opens up	Add it as a signal; refit before it affects scores
A health system announces a merger	The next news run moves its doctors up the timing list
The team has less capacity	Raise the score cut-off
The Company expands	Switch groups from city limits to drive time

Next 90 days

A pilot that produces real outcomes is the most valuable next step.



Pilot plan: three 30-day phases, a go/no-go check at each

If top-scored doctors don't book more meetings than the holdout by day 60, the signals need rethinking.

Assumptions and open questions

- "Will move" means within 24 months; a group is local to one metro and needs five or more.
- "Hospital-based" and "independent-style group" are inferred from CMS billing and group names.
- Open questions: team capacity per month, priority metros, and whether contract dates can be logged.

How I used AI

Claude Code wrote most of the code under my direction. Four AI reviewers tried to break it and every finding was fixed with a test. Inside the product, Claude reads the news. No patient data was used.